

Yutong Wang

CONTACT INFORMATION

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RESEARCH INTEREST

I am a first-year PhD student from the University of Sydney. My research interest mainly focuses on machine learning, especially **multi-modal learning, optimal transport, video understanding and generation.**

EDUCATION

PhD, Computer Science
The University of Sydney, NSW, Australia Sep. 2025–Present
Advisor: *Prof. Chang Xu*

MPhil, Computer Science
Beijing Institute of Technology, Beijing, China Sep. 2022–Jun. 2025
Advisors: *Prof. Dixin Luo, Prof. Hongteng Xu*

Bachelor, Computer Science
Beijing Institute of Technology, Beijing, China Sep. 2018–Jun. 2022

PROJECT EXPERIENCE

Music-guided Trailer generation [ACM MM'24, IJCAI'25]

- Proposed an **inverse partial optimal transport-based music-guided trailer generator**, comprising a conditional movie shot selector and a movie-music shot aligner trained via bi-level optimization.
- Constructed a new **publicly available comprehensive movie-trailer dataset (CMTD)** for movie trailer generation and future video understanding tasks.

Self-supervised Video summarization [ACM MM'23]

- Proposed a **unified self-supervised framework** to solve generic and instructional video summarization tasks, using inverse optimal transport to jointly learn optimal transport plans and a projection module to align textual and visuals.
- Proposed an **effective method for generating frame-level pseudo-significance scores** based on the optimal transport plans, which are then used to train a keyframe selector without needing explicit annotations.

Set-supervised Temporal action alignment [ACM MM'22]

- Proposed a novel computational optimal transport technique called **unbalanced spectral fused Gromov-Wasserstein (US-FGW)** and applied it to set-supervised temporal action alignment, complemented by a new contrastive learning paradigm.
- Proposed a **learning strategy to compute the US-FGW distance** by leveraging the Bregman alternating direction method of multipliers (B-ADMM) algorithm.

INTERNSHIP EXPERIENCE

- Shanghai AI Laboratory, Research Intern** [CVPR'26] Apr. 2025–Present
 - Explore the application of VLM and reinforcement learning to achieve **graphic design tasks**, such as poster design, cover design, and web design.
 - Explore optimal transport distillation techniques to **accelerate video generation**, have achieved acceleration for the VACE model, enabling efficient video

generation in just 4 steps. Also, explore content-adaptive block routing techniques for DiT depth pruning and spatial-temporal aware width pruning, which further nearly halves per-step computation for both I2V and T2V generation tasks.

Advisors: Xinyuan Chen, Yaohui Wang

2. **Ant Group**, AliPay, *Research Intern* [CVPR'25] Apr. 2024–Oct. 2024
 - Participate in the audio-driven portrait animation project EchoMimic. My research topic is **video face enhancement**, including blind face video restoration and de-flickering in real-world and AI-generated videos.
 - Proposed a novel **3DVQGAN-based video enhancement framework with a powerful discriminator** that can support effective compression.
 - Proposed **spatial-temporal codebooks** for video quantization and **marginal prior regularization** to mitigate the codebook collapse problem.

Advisors: Jiajiong Cao, Chenguang Ma

3. **VRC Inc.**, *Research Intern* Jul. 2023–Sep. 2023
 - The research topic is **stylized video generation** based on text descriptions or referenced images, focusing on VLOG videos.
 - Proposed a stylized video generation pipeline, that can achieve DreamBooth-based style transfer, SD-based quality enhancement, and video stability enhancement (de-flickering).

Advisors: Hongteng Xu, Yingdi Xie

PUBLICATION CONFERENCE

1. **Yutong Wang**, Haiyu Zhang, Tianfan Xue, Yu Qiao, Yaohui Wang, Chang Xu, Xinyuan Chen - VDOT: Efficient Unified Video Creation via Optimal Transport Distillation, IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026.
2. Sidan Zhu, **Yutong Wang**, Hongteng Xu, Dixin Luo - Weakly-Supervised Movie Trailer Generation Driven by Multi-Modal Semantic Consistency, The International Joint Conference on Artificial Intelligence (**IJCAI**), 2025.
3. **Yutong Wang**, Jiajie Teng, Jiajiong Cao, Yuming Li, Chenguang Ma, Hongteng Xu, Dixin Luo - Efficient Video Face Enhancement with Enhanced Spatial-Temporal Consistency, IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2025.
4. **Yutong Wang***, Sidan Zhu*, Hongteng Xu, Dixin Luo - An Inverse Partial Optimal Transport Framework for Music-guided Movie Trailer Generation, The 32nd ACM International Conference on Multimedia (**ACM MM**), 2024.
5. **Yutong Wang**, Hongteng Xu, Dixin Luo - Self-supervised Video Summarization Guided by Semantic Inverse Optimal Transport, The 31st ACM International Conference on Multimedia (**ACM MM**), 2023.
6. Dixin Luo, **Yutong Wang**, Angxiao Yue, Hongteng Xu - Weakly-Supervised Temporal Action Alignment Driven by Unbalanced Spectral Fused Gromov-Wasserstein Distance, The 30th ACM International Conference on Multimedia (**ACM MM**), 2022.

PREPRINT

1. **Yutong Wang**, Yunke Wang, Tianfan Xue, Yu Qiao, Yaohui Wang, Xinyuan Chen, Chang Xu - PARE: Pruning and Adaptive Routing for Efficient Video Generation, arXiv 2605.27336.
2. **Yutong Wang**, Yunke Wang, Xinyuan Chen, Chang Xu - BEAT: Rhythm-Elastic Alignment for Agentic Music-guided Movie Trailer Generation, arXiv 2605.27067.

* denotes Equal contribution.

PATENT

1. An audio-guided video generation method. CN Patent ZL 202411174397.6
2. A self-supervised method for video summarization. CN Patent ZL 202311104554.1.
3. A Retrieval Method. CN Patent ZL 202211404021.0.
4. A Video Understanding Method. CN Patent ZL 202211405957.5.

TEACHING EXPERIENCE

<i>Tutor</i> , The University of Sydney.	Feb. 2026–May. 2026
Course Name: COMP 5329 Deep Learning	
Instructor: <i>Prof.</i> Chang Xu	
<i>Teaching Assistant</i> , Beijing Institute of Technology.	Sep. 2022–Jun. 2024
Course Name: Programming with Python	
Instructor: <i>Prof.</i> Dixin Luo	

HONOR

USydIS Scholarship, The University of Sydney	2025
Outstanding Graduate, Beijing Institute of Technology	2025
National Scholarship, Ministry of Education of China	2024
Travel Grant Award of ACM MM, MM	2024
Travel Grant Award of ACM MM, SIGMM	2023
Merit Student, Beijing Institute of Technology	2023
Outstanding Graduate, Beijing Institute of Technology	2022

SELECTED TALKS

Efficient Unified Video Creation via Optimal Transport Distillation
Invited talk on Sydney AI Meetup, May 2026.

Generative AI with Diffusion Models
Invited lecturer at USYD GenAI Summer School, Dec 2025.

Efficient Video Face Enhancement with Enhanced Spatial-Temporal Consistency
Invited talk on AI TIME, May 2025.

SERVICE

Reviewer (Conference): ACM MM, ICCV, WACV, CVPR, BMVC, ECCV, NeurIPS
Reviewer (Journal): T-MLR, T-IP